

input material in the form of a filament which it melts and forces through a thinner nozzle (the printer head) and lays it on the printed model where needed. The printer head moves along the X-Y axis of the design.

The RepRap Project is quite famous and many variations exist now which were built upon the original RepRap. The RepRap can print thousands of components limited only by the designs available. It can print complex objects, including scientific equipment, which would normally require industrial infrastructure.

Currently, the RepRap can only replicate its plastic components but its future target, over its evolutionary variations, is to achieve 100 per cent self replication.

2. Eventorbot

The mission of the Kickstarter project ‘Eventorbot! open source 3d printer’ by Duy Dang was to build a rigid, low cost 3D printer and it managed to achieve just that. The Eventorbot has a rigid frame and is made up of a single, long square tube that’s bent into a ‘U’ shape. This structure allows the Eventorbot to maintain rigidity making it less affected by vibrations while printing. The hollow tube is around 2.5” of square steel frame which also enables it to conceal wiring, thereby avoiding a mess of wires and looking aesthetic at the same time. It almost looks like a professional 3D printer. Eventorbot also eliminates around 40 per cent of the parts and costs required to assemble other 3D printers. The best part: around 80 per cent of



the Eventorbot’s parts (around 37) are printable. The Eventorbot has a printing size of around 152mm x 152mm x 152 mm (the Eventorbot 1.2 has a build area of 8” x 10” x 6”). Its nozzle printing precision is between 0.1mm-0.5mm. The material cost for the Eventorbot is between \$300 to \$500 and the assembled cost is around \$800. The 3D design plans for objects are available for free on www.thingiverse.com and the Eventorbot YouTube channel has a detailed video collection on the 3D printer’s assembly.